Eustachian tube

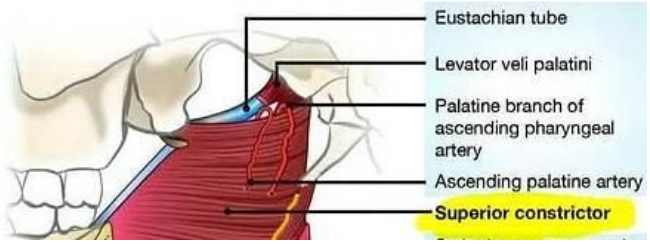
The tube that ventilates, drains and protects the middle ear: parts, dimensions, muscles, lining, and why children get more otitis media.

1. Parts and dimensions
2. Pharyngeal opening and relations
3. Muscles that open the tube
4. Lining, nerve and blood supply
5. Functions
6. Infant vs adult tube
7. In theatre
8. On imaging

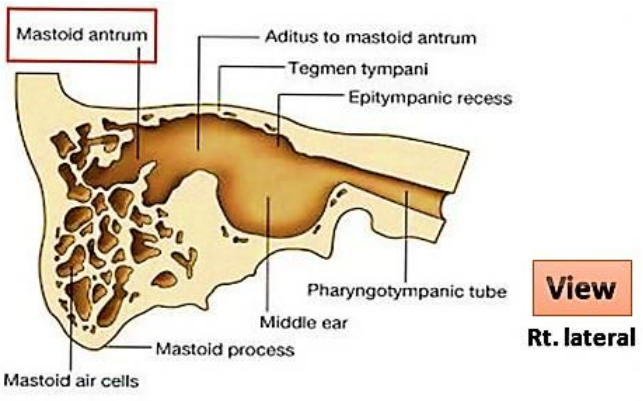
01 Parts and dimensions

- Connects the **anterior wall of the middle ear** (protympanum) with the **lateral wall of the nasopharynx**.
- Adult length about **36 mm**: **bony part (posterolateral third, about 12 mm)** in the petrous bone and **cartilaginous part (anteromedial two-thirds, about 24 mm)**.
- Runs **downward, forward and medially**, at about **45° to the sagittal plane** and about 30–40° to the horizontal in adults (often quoted simply as 45°).
- The **isthmus**, where the two parts meet, is the narrowest point (about 1 × 2 mm).
- The bony part is always open; the **cartilaginous part is closed at rest** and opens on swallowing, yawning and sneezing.
- The cartilage is hook-shaped in section: a **medial lamina** and a short **lateral lamina**, completed below and laterally by membrane.

> **CLINICAL PEARL** The tympanic opening is about 4 mm above the floor of the middle ear, so the middle ear drains uphill and relies on mucociliary clearance.



Eustachian tube with levator and tensor veli palatini
ENT Steps Anatomy, p. 204

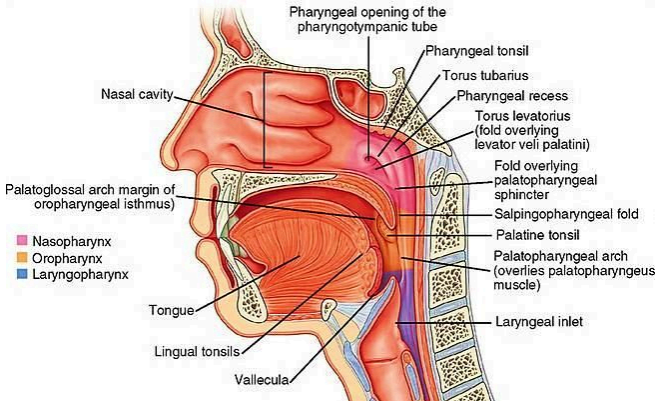


Mastoid antrum, aditus and epitympanic recess
ENT Steps Anatomy, p. 56

02 Pharyngeal opening and relations

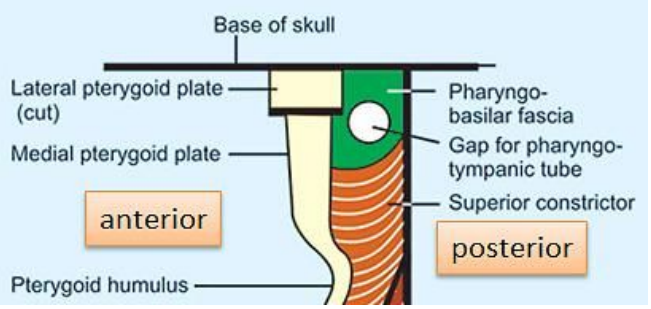
- Opens on the lateral wall of the nasopharynx about **1.25 cm behind the posterior end of the inferior turbinate**, at the level of the inferior meatus.
- Bounded above and behind by the **torus tubarius** (cartilage), with the **salpingopharyngeal fold** running down from it and the **fossa of Rosenmüller** behind.

- Enters the pharynx through the **sinus of Morgagni**, above the superior constrictor, with levator veli palatini.
- **Relations:** above, the skull base; below, superior constrictor; medially, origin of levator veli palatini; laterally, tensor veli palatini, the otic ganglion and the mandibular nerve.
- The **tensor tympani** lies in its own canal directly above the bony part.



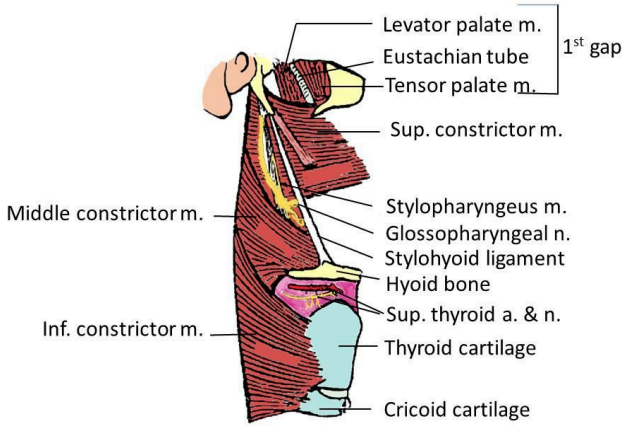
Lateral wall of the pharynx: torus tubarius, fossa of Rosenmuller

ENT Steps Anatomy, p. 203



Eustachian tube passing through the sinus of Morgagni

ENT Steps Anatomy, p. 204



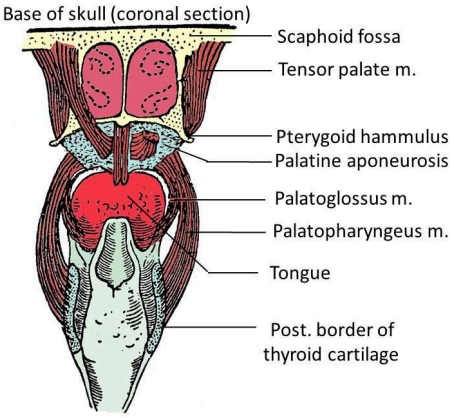
Gaps between the constrictors, lateral view

Dr. Sameh Doss handout, p. 167

03 Muscles that open the tube

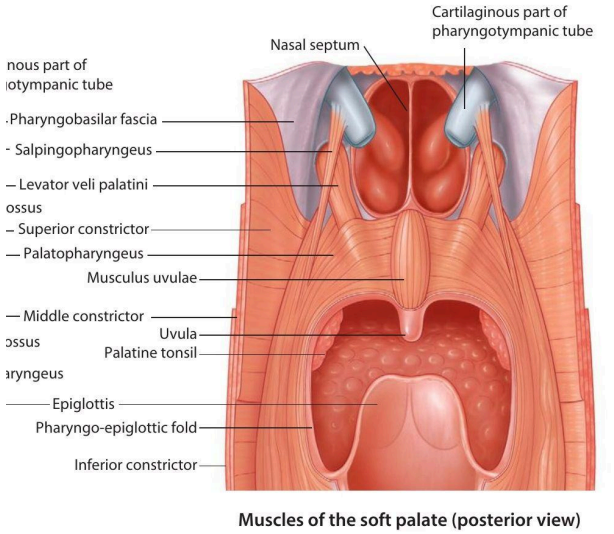
- **Tensor veli palatini (V3):** the main dilator. It arises from the lateral lamina and scaphoid fossa, hooks round the pterygoid hamulus and pulls the lateral wall away.
- **Levator veli palatini (X):** runs under the floor of the cartilaginous tube; its belly lifts the floor and rotates the medial lamina.
- **Salpingopharyngeus (X):** from the lower cartilage to the pharyngeal wall; minor role.
- **Ostmann's fat pad** lies laterally and helps keep the tube closed at rest.

> **CLINICAL PEARL** In cleft palate the tensor loses its anchor, so the tube does not open: most children with cleft palate have otitis media with effusion.



Muscles of the soft palate

Dr. Sameh Doss handout, p. 181



Muscles of the soft palate (posterior view)

Muscles of the soft palate (posterior view): tensor and levator veli palatini

Gray's Atlas of Anatomy, 2nd ed., p. 581

04 Lining, nerve and blood supply

- **Respiratory epithelium** (ciliated pseudostratified columnar) with goblet cells and seromucous glands, more glands toward the pharynx. Cilia beat toward the nasopharynx.
- **Lymphoid tissue (Gerlach's tubal tonsil)** at the pharyngeal end.
- **Nerves:** tympanic plexus (IX) for the bony part; pharyngeal branch of the pterygopalatine ganglion (V2) for the pharyngeal end.
- **Arteries:** ascending pharyngeal, middle meningeal (accessory meningeal), artery of the pterygoid canal. **Lymph:** retropharyngeal nodes.

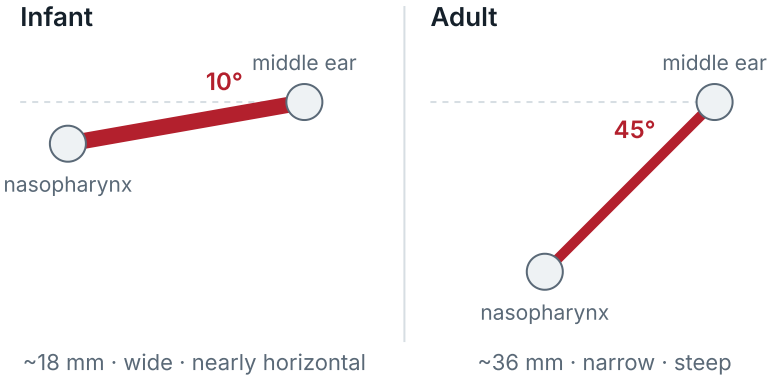
05 Functions

- **Ventilation:** equalises middle ear pressure with the atmosphere and replaces absorbed gas.
- **Clearance:** mucociliary and muscular drainage of middle ear secretions to the nasopharynx.
- **Protection:** the closed tube blocks nasopharyngeal sound and secretions (reflux); a cushion of air in the middle ear resists reflux.
- Normal tube opening needs an intact palate, working muscles and a non-inflamed lining.

> **CLINICAL PEARL** Patulous ET: the tube stays open (weight loss, pregnancy), causing autophony and hearing one's own breathing; symptoms ease when lying down.

06 Infant vs adult tube

- **Infant:** shorter (about 18 mm), **wider**, **more horizontal (about 10°)**, cartilage more floppy, opening at the level of the hard palate, and the tensor works less efficiently.
- **Adult:** about 36 mm, narrower, inclined about 30–45°, opening about 1 cm above the hard palate.
- The infant pattern lets nasopharyngeal secretions reflux into the middle ear and makes drainage poor, so **acute otitis media and OME peak in early childhood**.
- Adenoid hypertrophy adds obstruction and acts as a bacterial reservoir.



Short, wide, flat tube: easy reflux in, poor drainage out

Infant versus adult Eustachian tube. The angle and length change explain why otitis media peaks in early childhood and settles as the skull base grows.

07 Surgical anatomy: In theatre

- **Grommet insertion:** radial incision in the anteroinferior quadrant (away from the ossicles and with the best view).
- **Balloon dilation** of the cartilaginous ET for chronic obstructive dysfunction; stay medial in the tube and do not reach the bony part (the ICA canal lies close to it).
- **Adenoidectomy** for recurrent otitis media in children over about 2 years removes a bacterial reservoir next to the tube opening.

> CLINICAL PEARL The ICA lies just medial to the bony ET and may be dehiscent: never force an instrument into the bony isthmus.

08 Radiological anatomy: On imaging

- The ET is not routinely imaged; **CT** shows the bony part and a patulous tube (air-filled cartilaginous lumen).
- **MRI** of the nasopharynx is the key test when a unilateral middle ear effusion suggests a tumour blocking the tube.

T Tables

Eustachian tube: infant vs adult

	Infant	Adult
Length	~18 mm	~36 mm
Angle to horizontal	~10°	~30–45°
Width	Relatively wide	Narrow
Cartilage	Floppy, less dense	Firm
Tensor veli palatini efficiency	Poor	Good
Pharyngeal opening	Level of hard palate	~1 cm above hard palate

Explains the peak of otitis media in early childhood.

Eustachian tube: bony vs cartilaginous part

	Bony part	Cartilaginous part
Proportion	Posterolateral third (~12 mm)	Anteromedial two-thirds (~24 mm)
State at rest	Always open	Closed
Lining	Fewer glands, flatter	Ciliated with many glands
Relations	ICA medially, tensor tympani above	Tensor and levator veli palatini, otic ganglion, V3
Clinical	Isthmus at its junction; never force instruments	Target for balloon dilation

Tympanogram types

Type	Pattern	Typical cause
A	Normal peak at atmospheric pressure	Normal middle ear
As	Shallow peak	Otosclerosis, tympanosclerosis
Ad	Very tall peak	Ossicular discontinuity, flaccid TM
B	Flat	Effusion (normal volume), perforation or patent grommet (high volume), wax (low volume)
C	Peak at negative pressure	Eustachian tube dysfunction

Longitudinal muscles of the pharynx

Muscle	Origin	Insertion	Nerve	Action
Stylopharyngeus	Medial side of base of styloid process	Pharyngeal wall (passes between superior and middle constrictors)	Glossopharyngeal (IX) — its only muscle	Elevates pharynx
Salpingopharyngeus	Cartilaginous end of the Eustachian tube	Pharyngeal wall	Vagus (pharyngeal plexus)	Elevates pharynx
Palatopharyngeus	Upper surface of palatine aponeurosis	Pharyngeal wall and thyroid cartilage	Vagus (pharyngeal plexus)	Elevates pharynx; closes oropharyngeal isthmus

Gray's Atlas. All except stylopharyngeus are supplied by X through the pharyngeal plexus.

Muscles of the soft palate

Muscle	Origin	Insertion	Nerve	Action
Tensor veli palatini	Scaphoid fossa, spine of sphenoid, cartilaginous Eustachian tube	Palatine aponeurosis (tendon hooks round the hamulus)	V3 (via nerve to medial pterygoid)	Tenses palate; opens the Eustachian tube
Levator veli palatini	Petrous temporal bone in front of carotid canal; tube cartilage	Upper surface of palatine aponeurosis	Vagus (pharyngeal plexus)	Only elevator of the soft palate above neutral
Palatopharyngeus	Upper surface of palatine aponeurosis	Pharyngeal wall	Vagus (pharyngeal plexus)	Depresses palate; narrows isthmus; elevates pharynx
Palatoglossus	Under surface of palatine aponeurosis	Side of tongue	Vagus (pharyngeal plexus)	Depresses palate; elevates back of tongue

Muscle	Origin	Insertion	Nerve	Action
Musculus uvulae	Posterior nasal spine	Mucosa of uvula	Vagus (pharyngeal plexus)	Elevates and retracts uvula

Gray's Atlas. Tensor is the only palatal muscle not supplied by the pharyngeal plexus.

M Mnemonics

ET dimensions

36 = 12 + 24

Total 36 mm: 12 mm bony, 24 mm cartilaginous

Muscles opening the ET

TLS

Tensor veli palatini (main), Levator veli palatini, Salpingopharyngeus

Q Rapid recall

1. **Length of the adult Eustachian tube and its parts?** About 36 mm: bony third (12 mm) posterolateral, cartilaginous two-thirds (24 mm) anteromedial.
2. **Narrowest part of the Eustachian tube?** The isthmus, where the bony and cartilaginous parts meet.
3. **Main muscle that opens the Eustachian tube?** Tensor veli palatini (V3).
4. **Where does the ET open in the nasopharynx?** About 1.25 cm behind the posterior end of the inferior turbinate.
5. **Lining of the Eustachian tube?** Ciliated pseudostratified columnar epithelium with goblet cells and seromucous glands.
6. **Three functions of the Eustachian tube?** Ventilation (pressure equalisation), clearance (drainage), protection (from sound and reflux).
7. **Why do infants get more otitis media?** The ET is shorter, wider and more horizontal, with less efficient tensor action.
8. **Symptoms of a patulous Eustachian tube?** Autophony and hearing one's own breathing, eased by lying down or with a cold.
9. **Which state is the cartilaginous ET in at rest?** Closed; it opens on swallowing, yawning and sneezing.
10. **Nerve supply of the Eustachian tube?** Tympanic plexus (IX) and the pharyngeal branch of the pterygopalatine ganglion (V2).
11. **Which structure lies just medial to the bony ET?** The internal carotid artery in its canal.
12. **Where is a grommet usually inserted?** Anteroinferior quadrant of the pars tensa.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.